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Unique Pair Sum with Sets

Given a list of integers `nums` and an integer `target`, find all unique pairs of numbers in the list that add up to the `target`. Return the pairs as a list of tuples. Each pair should be unique, and the order of the numbers in the pairs does not matter.

You may assume that each input will have at least one valid pair, and the same number cannot be used twice in the same pair.

Constraints:

- `2 <= nums.length <= 10^4`
- `-10^9 <= nums[i] <= 10^9`
- `-10^9 <= target <= 10^9`

Sample Input and Output with Explanation:

Example 1:

Input: `nums = [2, 7, 11, 15], target = 9`

Output: `[(2, 7)]`

Explanation:

Only one pair `(2, 7)` adds up to 9.

Example 2:

Input: `nums = [3, 2, 4, 6, 1], target = 6`

Output: `[(2, 4)]`

Explanation:

The pairs `(2, 4)` add up to 6. The order of numbers in pairs does not matter.

Test Cases:

1. **Input:** `nums = [1, 2, 3, 4, 5, 6], target = 7`
Output: `[(1, 6), (2, 5), (3, 4)]`
2. **Input:** `nums = [4, 4, 2, 2], target = 6`
Output: `[(2, 4)]`
3. **Input:** `nums = [0, -1, 2, -3, 1], target = -1`
Output: `[(0, -1), (2, -3)]`
4. **Input:** `nums = [10, -10, 20, -20, 30], target = 10`
Output: `[(20, -10), (-20, 30)]`
5. **Input:** `nums = [5, 5, 5, 5], target = 10`
Output: `[(5, 5)]`

Common Elements in K Arrays

Problem Statement:

Given a list of lists of integers `arrays`, where each sublist is a list of integers, find all unique integers that appear in every sublist.

Constraints:

- `1 <= arrays.length <= 10^3`
- `1 <= arrays[i].length <= 10^3`
- `0 <= arrays[i][j] <= 10^9`

Sample Test Case 1:

Input: `arrays = [[2, 4, 6], [4, 6, 8], [6, 4, 10]]`

Output: `[4, 6]`

Explanation:

Both 4 and 6 are present in all three sublists. Hence, the common elements are `[4, 6]`.

Sample Test Case 2:

Input: `arrays = [[1, 2], [2, 3], [2, 4]]`

Output: `[2]`

Explanation:

Only 2 is present in all three sublists. Hence, the common element is `[2]`.

Test Cases:

1. **Input:** `arrays = [[1, 2, 3], [4, 5, 1], [1, 6, 7]]`
Output: `[1]`
2. **Input:** `arrays = [[1, 2, 3], [2, 3, 4], [3, 4, 5]]`
Output: `[3]`
3. **Input:** `arrays = [[10, 20], [20, 30], [20, 40]]`
Output: `[20]`
4. **Input:** `arrays = [[1, 1, 1], [1, 1, 1], [1, 1, 1]]`
Output: `[1]`
5. **Input:** `arrays = [[0, 1, 2], [1, 2, 3], [2, 3, 4]]`
Output: `[2]`

Movie Night Overlap

Scenario:

A group of friends is planning a movie night, but everyone has different tastes. Each friend provides a list of movies they are willing to watch. Your task is to find out which movies all friends agree upon, ensuring everyone will enjoy the movie night.

Problem Statement:

Given a list of lists `movies` where each sublist contains movies one friend is willing to watch, find all movies that appear in every friend's list.

Constraints:

- `1 <= movies.length <= 100`
- `1 <= movies[i].length <= 50`
- Movies are represented as strings with a length of up to 100 characters.

Test Cases:

1. **Input:** `movies = [["Inception", "Titanic"], ["Inception", "Avatar"], ["Inception", "Titanic", "Avatar"]]`
Output: `["Inception"]`
Explanation:
"Inception" is the only movie common in all friends' lists.
2. **Input:** `movies = [["Avengers", "Spiderman"], ["Avengers", "Batman"], ["Avengers", "Superman"]]`
Output: `["Avengers"]`
Explanation:
"Avengers" is the only movie all friends agree upon.

Additional Test Cases:

- **Input:** `movies = [["Harry Potter", "LOTR"], ["LOTR"], ["LOTR", "Star Wars"]]`
Output: `["LOTR"]`
- **Input:** `movies = [["Frozen", "Coco"], ["Frozen"], ["Frozen", "Moana"]]`
Output: `["Frozen"]`
- **Input:** `movies = [["Shrek", "Toy Story"], ["Shrek", "Toy Story"], ["Shrek"]]`
Output: `["Shrek"]`
- **Input:** `movies = [["A", "B", "C"], ["B", "C"], ["C"]]`
Output: `["C"]`
- **Input:** `movies = [["D"], ["D"], ["D"]]`
Output: `["D"]`

Conference Attendees

Scenario:

You are organizing a conference and want to identify attendees who have registered for all workshops. Each workshop has a list of registered participants.

Problem Statement:

Given a list of sets `workshops`, where each set contains the names of attendees registered for that workshop, find the names of attendees who registered for all workshops.

Constraints:

- `1 <= workshops.length <= 50`
- `1 <= len(workshops[i]) <= 100`
- Names are strings of up to 50 characters.

Test Cases:

1. **Input:**

```
workshops = [{"Alice", "Bob"}, {"Alice", "Charlie"}, {"Alice", "David"}]
```

Output:

```
["Alice"]
```

Explanation:

Alice is the only attendee who registered for all workshops.

2. **Input:**

```
workshops = [{"John", "Doe"}, {"John", "Smith"}, {"John", "Jane"}]
```

Output:

```
["John"]
```

Explanation:

John is the only attendee in all workshops.

Additional Test Cases:

- **Input:** `workshops = [{"Mark", "Tom"}, {"Mark"}, {"Mark", "Jerry"}]`
Output: `["Mark"]`
- **Input:** `workshops = [{"Sara"}, {"Sara", "Sam"}, {"Sara", "Tim"}]`
Output: `["Sara"]`
- **Input:** `workshops = [{"A", "B"}, {"B", "C"}, {"B"}]`
Output: `["B"]`
- **Input:** `workshops = [{"X"}, {"Y"}, {"Z"}]`
Output: `[]`
- **Input:** `workshops = [{"L", "M", "N"}, {"M", "N"}, {"M", "N"}]`
Output: `["M", "N"]`

Club Membership Overlap

Scenario:

Several clubs in a school have overlapping memberships. Given the membership lists of several clubs, determine which students are members of all the clubs.

Problem Statement:

Given a list of lists `clubs`, where each sublist contains names of students in that club, find the students who are members of every club.

Constraints:

- `1 <= clubs.length <= 30`
- `1 <= clubs[i].length <= 200`
- Names are strings with up to 100 characters.

Test Cases:

1. **Input:**

```
clubs = [["Alice", "Bob", "Charlie"], ["Alice", "David", "Bob"],  
         ["Bob", "Alice"]]
```

Output:

```
["Alice", "Bob"]
```

Explanation:

Alice and Bob are members of all clubs.

2. **Input:**

```
clubs = [["Tom", "Jerry"], ["Jerry", "Spike"], ["Jerry"]]
```

Output:

```
["Jerry"]
```

Explanation:

Jerry is the only member common to all clubs.

Additional Test Cases:

- **Input:** `clubs = [["Sara"], ["Sara", "Sam"], ["Sara"]]`
Output: `["Sara"]`
- **Input:** `clubs = [["A", "B"], ["B"], ["B", "C"]]`
Output: `["B"]`
- **Input:** `clubs = [["X"], ["Y"], ["Z"]]`
Output: `[]`
- **Input:** `clubs = [["Mark", "Tom"], ["Mark", "Jerry"], ["Mark"]]`
Output: `["Mark"]`
- **Input:** `clubs = [["Alice", "Bob"], ["Alice"], ["Bob"]]`
Output: `[]`

Shared Library Books

Scenario:

Multiple libraries in a city share a collection of books. You want to identify the books available in all libraries.

Problem Statement:

Given a list of sets `libraries`, where each set contains titles of books available in that library, return the titles of books that are available in every library.

Constraints:

- `1 <= libraries.length <= 20`
- `1 <= len(libraries[i]) <= 100`
- Book titles are strings with up to 150 characters.

Test Cases:

1. **Input:**

```
libraries = [{"Harry Potter", "LOTR"}, {"Harry Potter", "1984"}, {"Harry Potter", "Pride and Prejudice"}]
```

Output:

```
["Harry Potter"]
```

Explanation:

"Harry Potter" is the only book available in all libraries.

2. **Input:**

```
libraries = [{"Book A", "Book B"}, {"Book B", "Book C"}, {"Book B"}]
```

Output:

```
["Book B"]
```

Explanation:

"Book B" is available in all libraries.

Additional Test Cases:

- **Input:** `libraries = [{"A", "B"}, {"B"}, {"B", "C"}]`
Output: `["B"]`
- **Input:** `libraries = [{"X"}, {"Y"}, {"Z"}]`
Output: `[]`
- **Input:** `libraries = [{"Harry", "Tom"}, {"Harry", "Jerry"}, {"Harry"}]`
Output: `["Harry"]`
- **Input:** `libraries = [{"A"}, {"A", "B"}, {"A", "C"}]`
Output: `["A"]`
- **Input:** `libraries = [{"Book 1", "Book 2"}, {"Book 1"}, {"Book 2"}]`
Output: `[]`

Festival Food Stalls

Scenario:

During a festival, various food stalls offer different dishes. You want to find out which dishes are common across all stalls.

Problem Statement:

Given a list of sets `food_stalls`, where each set contains dishes offered by a stall, return a list of dishes that are available at every stall.

Constraints:

- `1 <= food_stalls.length <= 15`
- `1 <= len(food_stalls[i]) <= 50`
- Dishes are strings with up to 100 characters.

Test Cases:

1. **Input:** `food_stalls = [{"Pizza", "Burger"}, {"Pizza", "Pasta"}, {"Pizza", "Salad"}]`
Output: `["Pizza"]`
Explanation:
"Pizza" is the only dish common across all stalls.
2. **Input:** `food_stalls = [{"Curry", "Rice"}, {"Curry", "Naan"}, {"Curry"}]`
Output: `["Curry"]`
Explanation:
"Curry" is the only common dish.

Additional Test Cases:

- **Input:** `food_stalls = [{"Ice Cream"}, {"Ice Cream", "Cake"}, {"Ice Cream", "Pudding"}]`
Output: `["Ice Cream"]`
- **Input:** `food_stalls = [{"A"}, {"A", "B"}, {"A"}]`
Output: `["A"]`
- **Input:** `food_stalls = [{"Soup"}, {"Soup"}, {"Soup"}]`
Output: `["Soup"]`
- **Input:** `food_stalls = [{"X", "Y"}, {"Y"}, {"Y", "Z"}]`
Output: `["Y"]`
- **Input:** `food_stalls = [{"Pizza"}, {"Burger"}, {"Salad"}]`
Output: `[]`